



GEOGRAPHY

6890/01

Paper 1 Geographical Themes

For examination from 2027

Total marks: 75

Specimen

MARK SCHEME

Note to examiners

Levels of response marking is used for section (c) of each question. Thus, it is the quality of the response that determines which level an answer has achieved rather than the quantity of statements contained within it. However, once assigned to a level the mark achieved within that level is determined by the number of points made. Levels 1 and 2 are distinguished by whether statements are simple (level 1) or developed/elaborated (level 2). A candidate can immediately enter L2 by making developed points without making any L1 statements. In order to achieve L3 a candidate must have already reached the top end of L2. In addition, their answer should have a clear example and if the answer is place specific as well (7 marks). Highlight place specific detail.

Summary

Level 1 (1 to 3 marks):

- 1 simple statement (1 mark)
- 2 simple statements (2 marks)
- 3 simple statements (3 marks)

Level 2 (4 to 6 marks):

- 1 developed statement (4 marks)
- 2 developed statements (5 marks)
- 3 or more developed statements with e.g. (6 marks)

Level 3 (7 marks):

- 3 or more developed statements + named example with at least one piece of place specific detail.

- 1 (a) (i) Define a *river channel*
path or course through which a river flows 1 @ 1 = 1
- (ii) Study Fig. 1, which shows two sketch maps of the same river, Sketch **A** and Sketch **B**. Sketch **B** was drawn several years after Sketch **A**.

State the river processes that have taken place at **X** and **Y**.

X – Erosion

Y – Deposition

2 @ 1 = 2

- (iii) Describe **three** changes that have taken place in Sketch **B** since Sketch **A** was drawn.

- **More erosion at X**
- **More deposition at Y**
- **Widening of river channel**
- **Road has collapsed/ river has undercut road**

3 @ 1 = 3

- (iv) Describe **four** reasons why the rate of river erosion varies.

- **Gradient/slope of the river**
- **Velocity/speed of the river**
- **Seasonal flow/regime of the river**
- **Volume of water in the river**
- **Size of the river channel**
- **Rock structure in which river flowing**

4 @ 1 = 4

- (b) Study Fig. 2 (Insert), which shows part of a river.

- (i) Identify **four** river landforms shown in Fig. 2.

- **Flood plain**
- **Bluffs**
- **Levees**
- **Oxbow lakes**
- **River cliff**
- **Point bars**
- **Meanders**

4 @ 1 = 4

- (ii) Explain **two** negative impacts rivers may have on the environment.

- **Destruction of vegetation (cover) – during flooding the river overflows its banks and sweeps away trees and grass**
- **Loss of habitats – siltation may fill up dams/ ponds which are habitats to many species**
- **Soil erosion – the banks and bed are eroded by the river**
- **Destruction of ecosystems – due to loss of habitats ecosystems are disrupted**

2 @ 2 = 4

(c) Rivers offer some opportunities to people. Using an example of an area you have studied, suggest **three** opportunities presented by rivers.

- **water supply – for domestic use/ industrial use/irrigation**
- **transport – some rivers are navigable allowing ships to sail**
- **source of food – as fishing is practiced in rivers**
- **building material – in the form of sand/stone/gravel for building houses, etc.**
- **Fertile soils – in flood plains allowing crop growing**
- **tourist attraction – as some rivers have beautiful sceneries such as waterfall/gorges, etc.**
- **development of harbours – some river mouths have deep sheltered waters allowing development of sea-ports**
- **minerals – obtained in alluvial deposits leading to opening of mines and employment opportunities**
- **rivers act as natural boundaries between countries**
- **power generation – river water is used to generate hydro-electricity**

3 @ 2 plus example = 7

2 (a) Study Photographs **A**, **B**, **C** and **D** (Insert), which show causes of climate change.

(i) Define *climate change*.

sudden change in temperature and rainfall patterns

1 @ 1 = 1

(ii) Identify two gases released from the event shown in Photograph **A** that can cause climate change. Choose your answers from the gases listed in the box below.

- **carbon dioxide**
- **sulfur dioxide**

2 @ 1 = 2

(iii) State the cause of climate change shown in each of the Photographs **B**, **C** and **D**.

B – deforestation

C – gases from air conditioners

D – industry

3 @ 1 = 3

(iv) Describe **four** impacts of climate change.

- **changing rain and snow patterns**
- **change in animal migration and life cycle**
- **higher temperatures and more heat waves**
- **damaged corals**
- **rising sea-level**
- **more droughts and wildfires**
- **warmer oceans**
- **less snow and ice**
- **changes in plant lifecycles**
- **thawing permafrost**

4 @ 1 = 4

(b) Study Fig. 3, which shows how the greenhouse effect works.

(i) Using Fig. 3, explain how the greenhouse effect works.

- **incoming solar (shortwave) radiation/heat from the sun**
- **(shortwave radiation) heat from the sun passes through the atmosphere**
- **some of the heat is absorbed by the land and oceans**
- **the heat radiated back from the surface**
- **(longwave) radiated heat is trapped by a layer of gases in the atmosphere**

4 @ 1 = 4

(ii) Explain **two** ways in which climate change can be reduced.

- **use of hybrid cars – which emit less carbon dioxide**
- **use other sources of energy like hydro, wind and geo-thermal – which are less pollutant**
- **preserve natural vegetation – that act as carbon gas sinks**
- **plant trees – that will act as carbon gas sinks**
- **encourage use of public transport – to reduce number of cars on roads**
- **educate people of climate change and how it can be mitigated**

2 @ 2 = 4

(c) Certain strategies need to be applied to adapt to climate change. For a country you have studied, suggest **three** ways of adapting to an environment that has been changed by climate change.

- **harvest water – and store it in tanks for use in times of drought**
- **government should put policies - to reduce climate change such as use of unleaded fuel**
- **raising awareness – through holding climate change workshops**
- **install early warning systems – for people to take safety precautions if extreme events will occur such as cyclones**
- **avoid building on coastlines – to prevent flooding**
- **build flood proof houses on stilts – to be safe from floods**

3 @ 2 plus example = 7

3 (a) Study Fig. 4, which is a plan of Matsapha Industrial Site in Eswatini.

(i) Define an *industry*.

- **an activity of economic value**

1 @1 = 1

(ii) Using Fig. 4, give any **two** examples of secondary industries found at Matsapha Industrial Site.

- **Engineering works**
- **Afrisam cement**
- **Eswatini beverages**
- **Eswatini wire**

2 @ 1 = 2

(iii) Using Fig. 4 and any other information, describe **three** disadvantages of the location of Matsapha Industrial Site for the surrounding areas.

- **Air, water, visual, sound pollution** (max 2)
- **Water shortage**
- **Increase in rates**
- **Overcrowding of surrounding residential areas as people flock for employment**
- **Loss of arable land(land reserved for industry)**

3 @ 1 = 3

(iv) Describe **four** positive impacts of Multi-National Companies in countries such as Eswatini.

- **offer employment to many people in countries where they operate**
- **local work force get guaranteed income**
- **bring income to country**
- **government get taxes**
- **improves level of education and technical skills of the people**
- **lead to improvement of infrastructure like roads, railways, etc.**
- **widen economic base of country**

4 @ 1 = 4

(b) Study Photograph E (Insert), which shows a craft industry in Eswatini.

(i) Describe **four** features of the craft industry shown in Photograph E.

- **small industry/cottage industry**
- **handmade products**
- **few labourers**
- **mainly women labour**
- **simple tools**
- **natural inputs/grass**

4 @ 1 = 4

(ii) Explain **two ways** in which craft industries are important for Eswatini.

- **expose and make use of the traditional skills that the people have – and people earn money in return**
- **provide self-employment to rural people – improving the standard of living**
- **reduces the rate of rural-urban migration – reducing negative impacts thereof**
- **make use of the locally obtained raw materials – cheaper hence more profits for the craft makers**
- **brings income to the rural people – may in turn improve their communities**
- **aids in the tourism industry – generating foreign revenue for the country**

2 @ 2 = 4

(c) High technology industries are usually found in MEDCs. For a country you have studied, suggest **three** factors that influence the location of high technology industries.

- **near universities and other research departments - for skilled labour**
- **access to developed international transport systems (airports) – for quick transportation**
- **next to other related firms – to share ideas and information**
- **in areas where unemployment is high – to get labour**
- **in pleasant areas with attractive surroundings.**

3 @ 2 plus example = 7

4 (a) Study Fig. 5A which shows sources of global energy and Figs. 5B and 5C which show sources of energy in South America and Europe.

(i) Circle **one** example of a non-renewable source of energy from the box below.

Natural gas	Solar	Hydro energy
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circle Natural gas 1 @ 1 = 1

(ii) Using Fig. 5A and Fig. 5C, identify the source of energy that is:

A Most used globally – oil

B Least used in Europe – hydro energy

2 @ 1 = 2

(iii) Using Fig. 5B and Fig. C, state **three** differences between the sources of energy used in South America and Europe.

- **oil used in Europe, none in South America**
- **more HEP used in South America than in Europe**
- **more nuclear power used in Europe than South America**
- **natural gas more in South America than in Europe**
- **coal used in Europe , none in South America**

3 @ 1 = 3

(iv) Describe **four** negative impacts of the use of fossil fuels such as coal and oil on the environment.

- **air pollution**
- **acid rain**
- **damage to plant life and food chains**
- **global warming**
- **rise in sea-levels**

4 @ 1 = 4

(b) Study Photograph F (Insert), which shows a wind farm.

(i) Using Photograph F and any other information, describe how electricity is generated by wind.

- wind makes the blades on a wind turbine spin,
- this creates kinetic energy
- the kinetic energy makes the shaft in the nacelle to turn
- a generator in the nacelle converts this kinetic energy into electrical energy.

4 @ 1 = 4

(ii) Explain **two** positive impacts of generating power using renewable resources such as wind.

Pollution reduction – as there are less emissions

Cheaper to sustain – less expenses for raw material/ free raw material

Inexhaustible – the energy source is available naturally

2 @ 2 = 4

(c) Many LEDCs are faced with a shortage of power supply. For an area or country you have studied, suggest **three** negative impacts of a shortage of power.

Closure of factories/industries leading to – loss of jobs/ decline in revenue

Taps run dry – leading to sewage problems

Disrupts domestic life – e.g. cooking schedule disturbed/ entertainment

Disrupts the flow of traffic – which disrupts communications.

Decline in production – as mines are closed

Health systems are disturbed – leading to loss of life

3 @ 2 plus example = 7

5 (a) Study Fig. 6, which shows three settlement patterns **A**, **B** and **C**.

(i) Define a *settlement pattern*.

shape formed by settlements in an area

1 @ 1 = 1

(ii) Identify the **two** settlement patterns **A** and **B**.

A – linear/line

B – nucleated/clustered

2 @ 1 = 2

(iii) State factors that might have influenced the location and development of each settlement shown in Fig. 6.

A – plantation

B – sea

C – route focus/roads

3 @ 1 = 3

(iv) Give **four** factors which determine the position of a settlement in the settlement hierarchy.

- **population of the settlement**
- **the services provided by the settlement**
- **the spheres of influence of the settlement**
- **the size of the settlement**

4 @ 1 = 4

(b) Study Fig. 7 which is a simple map of Mbabane, an urban area in Eswatini.

(i) Using Fig. 7, identify **four** services that are provided in Mbabane.

- **banking**
- **administration**
- **religion**
- **accommodation**
- **entertainment**
- **marketing**
- **education**

4 @ 1 = 4

(i) Explain **two** negative impacts of a large number of people coming to live in a city such as Mbabane.

- **development of shanty towns – due to high cost of housing or lack of housing**
- **high cost of living – leading to lack of access to services**
- **strain on resources such as water supply – resulting to spread of diseases**
- **increased pollution – which may lead to respiratory disease**
- **poverty – leading to crime, beggars and street kids**

2 @ 2 = 4

(c) Traffic congestion is a problem in cities in both LEDCs and MEDCs. For a country you have studied, suggest **three** solutions that may reduce traffic congestion.

- **increase public transport – prevent use of private cars**
- **by-pass roads – to ease congestion at the city centre**
- **paid parking – to reduce parking hours by shoppers**
- **flexi-time – to reduce congestion during rush hours**
- **highways/flyover roads – allow free flow of traffic at high speeds**
- **widen streets/multi-lane streets – to accommodate high traffic volume**
- **car parks – to avoid street parking**

3 @ 2 plus example = 7

6 (a) Study Fig. 8, which shows the population structure of an LEDC in Africa.

(i) Define *population structure*.

- **shape/make-up/composition of the population according to age and gender/sex**

1 @ 1 = 1

(ii) Describe **two** features of the population structure of the LEDC shown in Fig. 8.

- many young dependents/high population of children
 - less old dependents/fewer aging population
 - economically active fewer than young dependents/decrease in population as age increases
 - high dependency ratio
 - low life expectancy
- 2 @ 1 = 2

(ii) Suggest **three** reasons for the population structure shown in Fig. 8.

- Diseases
 - lack of clean water
 - poor sanitation
 - lack of food
 - high birth rate
 - lack of family planning education
 - culture/religion promote having many children
- 3 @ 1 = 3

(iii) State how each of the following may cause changes to the population structure of a country:

Decline in infant mortality

Increased population of young aged

Decline in birth rate

Decrease in population of young aged

Increase in emigration (migration out of a country)

Decline in population of working /adult age group

Pandemics such as HIV

Increase in number of dependents

4 @ 1 = 4

(b) Study Fig. 9, which shows the percentage of people over the age of 60 years in an LEDC and an MEDC between 1990 and 2025 (projected).

(i) Compare the percentage of the population of people over 60 years between the LEDC and the MEDC between 2010 and 2025. Use statistics from Fig. 9 to support your answer.

- The percentage population of 60 years and above in MEDCs is higher than that of LEDCs
- The percentage of MEDC's population of 60 years and above increases more rapidly than LEDC's population
- 2010 MEDC's population of 60 years and above is around 7.6% while LEDC's is 4.6%
- 2020 MEDC's population of 60 years and above is 19% while LEDC's is 8%

- **2025 MEDC's population of 60 years and above is projected to be 29.5% while LEDC's will be 10%**

allow any comparison by years

4 @ 1 = 4

(i) Explain **two** reasons for the rapid increase in the population of people over 60 years in MEDCs.

- **better treatment of diseases – few will die**
- **more health care staff – to take care of the sick**
- **more medicines/drugs for vaccination – to prevent them from getting diseases**
- **more services for the elderly – to cater for their needs**
- **investment in old age care homes – to take good care of them**
- **improved food supply – ensures they are not starved**
- **better sanitation – improves their health**

2 @ 2 = 4

(c) Pandemics such as HIV/AIDS have an impact in both MEDCs and LEDCs.

For a country you have studied, suggest **three** efforts that can address the negative impact of HIV/AIDS.

- **education/awareness – community and media programmes where the public is taught about means of prevention, minimising spread**
- **free/subsidised treatment/prevention – like PMTCT, PrEP, PEP, ARVs to ensure wider population coverage, minimising spread and AIDS victims**
- **anti-discrimination/stigma free healthcare – national health strategies to provide non-judgemental healthcare thus encouraging early detection and widespread support**
- **international cooperation – governments cooperating with other nations or initiatives such as PEPFAR that provide funding for treatment or prevention programmes**

3 @ 2 plus example = 7